MILITARY SPECIFICATION
CARTRIDGE, GRENADE: 5.56MM, XM195

1. SCOPE

1.1 This specification covers a cartridge intended for propelling grenades from 5.56MM rifles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

C10534926	- Cartridge, Grenade: 5.56MM, XM195
C7643674	- Classification of Cartridge Case Defects
F10542291	- Packing and Marking, Cartridges, Grenade, 5.56MM, XM195, Cartons; Box, Ammunition, M2A1; Box, Wirebound
IEL-10534926	- Inspection Equipment List for Cartridge, Grenade, 5.56MM, XM195

PUBLICATIONS

Munitions Command

FSC 1330

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AMSMU-P715-501FA1 - Ammunition Ballistic Acceptance Test Methods, Test Procedures for 5.56mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with drawing C10534926, referenced specifications and the following:

3.2 First article testing.- Requirements for the submission of first article samples by the contractor shall be specified in the contract.

3.3 Residual stress.- The sample cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproof.- The sample cartridges shall not release more than one bubble of air when subjected to a pressure of 5 pounds per square inch (PSI) below atmospheric pressure and held at that pressure for 15 seconds.

3.5 Velocity.- The sample cartridges fired in a horizontal position and at the indicated temperatures shall propel grenades weighing 1.56 pounds plus or minus 0.01 pounds (1.56 ± .01) at velocities specified below. The velocity shall be measured at a point 5 feet, 6 inches beyond the forward end of the grenade when fully positioned for launching.

	Instrumental Velocity Ft per Second (ft/sec)	
Degree Fahrenheit (F)	Minimum Individual	Maximum Individual
65 to 75	140	165
125 to 130	140	165
-40 to -45	140	165

3.6 Function and Casualty.- The velocity test and function and casualty test are fired simultaneously. The sample cartridges shall function without casualty at temperatures of minus 40 degrees to minus 45 degrees F. at ambient temperature and at plus 125 degrees to plus 130 degrees F.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

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3.7.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.7.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discoloration, dirt, oil and smears of lacquer.

3.8 Primer sensitivity.- The energy imparted by a 3.94 ± .02 ounce steel ball falling 12 inches shall cause initiation of the primer. The energy imparted by a 3.94 ± .02 ounce steel ball falling 3 inches shall not cause initiation of the primer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 550 cartridges and 600 primed cartridge cases. The sample shall be manufacutred using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

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4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the manufacturer's identification as assigned by the procuring activity. Each lot shall be further identified by a federal stock number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1.	Discolored, dirty, oily			X	
2.	Corroded or stained, if etched		X		
	Case				
4.	Round head		X		
5.	Dent				X
6.	Split case	X			
7.	Perforated case	X			
8.	Draw Scratch				X
9.	Scratch			X	
10.	Beveled underside of head		X		
12.	Scaly metal				X
13.	No chamfer on head (rim)		X		
14.	Fold			X	
15.	Wrinkle			X	
17.	Bulge			X	
18.	Illegible or missing head stamp			X	
19.	Defective head			X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Primer				
32.	No primer	X			
33.	Cocked primer	X			
34.	Inverted primer	X			
35.	Loose primer		X		
36.	Nicked or dented primer			X	
37.	No waterproofing material (primer pocket joint)			X	
38.	Defective crimp			X	
	Mouth				
39.	Missing wad	X			
40.	Wad lacquer missing or inadequate		X		
42.	Mouth crimp missing or incorrect			X	
42a.	Opening between wad and case wall		X		
	Gaging				
43.	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)		X		
44.	Diameter of extractor groove			X	
45.	Diameter of head			X	
46.	Thickness of head, max.			X	
47.	Thickness of head, min.		X		
48.	Depth of primer		X		
	Weighing				
49.	Weight, min.	X			

1/ Refer to MIL-STD-636 for visual standards for defects 1 through 38. While no photographs of grenade cartridges are included in this Standard, the cartridge case defects illustrated in the Nato Caliber 7.62mm section can be used. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot.

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The cartridges shall be thoroughly mixed before being divided into the samples for the various tests.

Table II

Number of Cartridges

Test	Temperature F. Ambient Temp.	65° to 75°	-40° to -45°	125° to 130°	Requirement Paragraph
Residual Stress (Mercurous Nitrate) 1/	50	--	--	--	3.3
Waterproof 2/	20	--	--	--	3.4
Velocity	--	25 3/	25 3/	25 3/	3.5
Function and Casualty 5/					3.6
Primer Sensitivity 6/					3.8

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

3/ The variables sampling plan of MIL-STD-414, example C-4 for an AQL of 0.25 percent for the lower limit and an AQL of .65 percent for the upper limit shall be used for each sample of 25 cartridges. Failure of any cartridge to comply with the requirements of 3.5 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot. No retest permitted.

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4/ The lot shall be rejected when function and casualty defects noted in the firing of the velocity test exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the initial sample exceeds the acceptance number for the initial sample of Table III but is equal to or less than the acceptance number for the cumulative sample (Table III), a second sample consisting of double the quantity specified for the velocity test, shall be fired in the weapon specified therefor. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If in testing the second sample, defects other than those for which the second sample is being tested, should occur to the extent that they exceed the acceptance numbers for the cumulative sample, the lot shall be rejected.

5/ Prior to assembly into cartridges, a sample of primers from each primer lot shall be tested for sensitivity. The test sample shall consist of the number of primers required for performance of a complete run-down test wherein fifty primers are tested at each height. If the average critical height ($\bar{H}$)plus three standard deviations (3 sigma) exceeds 12 inches, or if the average critical height ($\bar{H}$) minus three standard deviations (3 sigma) is less than 3 inches, the lot of primers shall be rejected subject to performance of a second complete run-down test wherein one-hundred primers are tested at each height. Failure of the second sample to comply with the above criteria shall be cause for rejection of the lot of primers.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative First and Second Sample
1. Misfire	0	1
2. Primer Lead		
a. Perforation in firing pin indent in primer cup	1	5
b. Escape of gas through primer cup other than (a)	0	2
c. Escape of gas around primer cup 1/	1	2
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	1	2

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Table III (Cont'd)

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative First and Second Sample
3. Case Casualties		
a. Longitudinal split 3/		
(1) Body (J)	1	2
(2) Body (K)	0	2
(3) To head (L)	0	1
(4) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial, body (J)	0	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
4. Failure to extract	0	1

1/ Gas escapes around more than 50 percent of periphery of cup.

2/ No penalty shall be charged against a fired case with mouth partially or completely burned. If, however, such fired cartridge case shows the occurrence of some other firing casualty, exclusive of failure of rosebud crimp to open, such other casualty shall be penalized in accordance with Table III.

3/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a dummy, blank or any bullet type cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed in IEL-10534926.

4.4 Test methods and procedures.

4.4.1 Residual stress (Mercurous Nitrate).- The test shall be conducted in accordance with AMSMU-P715-501FA1.

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4.4.2 Waterproof.- The test shall be conducted in accordance with AMSMU-P715-505FA1.

4.4.3 Velocity.- The test shall be conducted in accordance with Appendix "A".

4.4.4 Function and casualty.- Observation for firing defects listed in Table III shall be made during the firing of the velocity test. Retest shall be fired in accordance with the procedure prescribed for the velocity test in Appendix "A", except that all reference pertaining to obtaining of velocity data shall be disregarded.

4.4.5 Temperature tests.- The test shall be conducted in accordance with Appendix "B".

4.4.6 Primer sensitivity.- After removal of propellant from the cartridge, the method and procedure for conducting the test shall be in accordance with AMSMU-P715-501FA1.

4.4.7 Defect penalty.- If, in the velocity test, the occurrence of a firing defect prevents the obtaining of a reliable result, an additional shot shall be fired. The velocity test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. (Worldwide shipment) The cartridges shall be packed in accordance with drawing F10542291.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection for each lot of ammunition presented to the government.

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6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodian:

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Preparing Activity:

Army-MU

Project No. 1330-A158

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APPENDIX "A"
Velocity, Function and Casualty Tests

1.0 PURPOSE

The Velocity, and Function and Casualty tests are fired simultaneously. The purpose of the Velocity is to determine if the average velocity of rifle-grenades, when propelled by the grenade cartridges, complies with the velocity requirement contained in Section 3.

The purpose of the Function and Casualty test is to demonstrate, by firing in weapons of representative type, whether or not the ammunition undergoing acceptance can be expected to perform satisfactorily in the service weapon for which it has been designed.

Casualties and malfunctions can be caused either by the ammunition or by the weapon in which it is fired, so that, to a certain extent, these two factors are inter-dependent. A faulty or poorly adjusted weapon can cause casualties in normal ammunition, but if the weapon be in proper condition when casualties are encountered, then the fault lies with the ammunition.

2.0 EQUIPMENT

2.1 Equipment listed in the Velocity section of the appropriate Inspection Equipment List shall be used.

2.2 The firing range shall be arranged as shown on Chart #1, at the end of this section. The maximum distance between the lumiline screens shall be ten feet.

2.3 Pier or mount of solid construction on which the test fixture assembly shall be mounted.

3.0 TEST PROCEDURE

3.1 Pre-firing (Preparative for test).

3.1.1 Weapons shall be of the latest design or most recent issue. The use of weapons or parts of earlier design is sometimes permitted for special purposes.

3.1.2 Prior to assembling the velocity test weapon in the test fixture, the applicable magazine (empty) shall be inserted in the weapon; bolt shall be in open position and the safety in "safe" position.

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3.1.3 The test weapon shall be assembled in the test fixture on the mount and sighted in on the recovery container down-range. Weapon shall be sighted-in for each cartridge to be fired.

3.1.4 The test cartridges shall be examined for obvious defects. If visual defects are found, the defective cartridge(s) shall be replaced and the defects shall be noted on the test sheet form.

3.1.5 The controlled temperature containers shall be brought to the specified temperatures before the test cartridges are placed in their respective containers. The test cartridges shall be brought to a temperature of $60^{\circ}F$ to $80^{\circ}F$ prior to being placed in their respective controlled temperature containers.

3.1.5.1 The required number of test cartridges shall be placed in a vertical position, primer-end down, in recessed holding blocks. The cartridges shall be placed in the controlled-temperature containers in such a manner that all cartridges are subjected to the specified temperature for the length of time prescribed in the item specification. Cartridges to be fired at $70^{\circ}F \pm 5^{\circ}F$ shall be stored in a container or room maintained at a temperature of $70^{\circ} \pm 2^{\circ}F$ with a relative humidity of 60 ± 5 percent for the time specified. The container or room shall be of sufficient capacity to allow free circulation of air.

3.1.6 Lumiline screens are checked for position. It is of the utmost importance that the lumiline screens be placed in their proper positions, measurements must be accurate. The lumiline screen nearest the muzzle of the velocity weapon shall be of blast proof construction. Range shall be equipped with a suitable recovery container filled with cotton waste or equivalent, for stopping and preserving a propelled proof-slug, so that recovery can be made. The recovery container should be located approximately fifteen feet from the muzzle of the velocity weapon, in the line of fire.

3.2 During Firing

3.2.1 When the test is being conducted, the chamber of the weapon shall be empty, the bolt in the open position and the safety selector placed in the "safe" position, until ready to fire. This precaution is necessary because the technician is required to place himself between the muzzle of the weapon and the recovery container during certain operations.

3.2.2 The proof-slug(s) is/are placed at a point convenient to the technician. Three warming (fouling) shots having components of the same type as the test cartridges are fired and velocity readings of the proof-slugs are recorded by the chronographer to assure that the equipment is functioning properly.

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3.2.3 The recessed holding block containing the test cartridges (reference cartridges are not fired) is removed from the controlled temperature container or room and placed at a point convenient to the technician, provided the temperature of the firing room is 70°F ± 5°F, otherwise the test cartridges shall be placed in an insulated box which has been conditioned at 70°F ± 5°F, and the box placed at a point convenient to the technician. Cartridges are then removed singly from the insulated box and fired. If an insulated box is not available then the cartridges shall be removed singly from the controlled temperature container or room and fired.

3.2.3.1 When firing cartridges which have been conditioned at either high or low temperature, the cartridges shall be placed in an insulated box which has also been conditioned at the specified temperature. No more than three cartridges shall be placed in the insulated box, at any one time. The insulated box containing the cartridges shall be placed at a point convenient to the technician; the cartridges are removed singly from the insulated box and fired. If an insulated box is not available, the cartridges shall be removed singly from the controlled-temperature container or room and fired.

3.2.4 A proof-slug is fully seated on the flash suppressor on the muzzle of the weapon.

3.2.5 In order to simulate actual combat conditions the cartridges are not rotated prior to insertion into the chamber of the test weapon; however, the manner of handling the ammunition from cartridge to cartridge shall be reasonably uniform.

3.2.5.1 As an extra safety precaution the charging handle shall be pulled fully to the rear and held in this position, using the left hand. Using the right hand, insert the cartridge into the chamber of the weapon. Release changing handle when right hand is clear of weapon.

3.2.6 The bolt is then released by depressing the upper portion of the bolt catch with the thumb of the left hand allowing the action to close. Do not "ride" the charging handle forward with the right hand. If the charging handle is eased forward from the open position, the bolt may fail to lock and the rifle will not fire (When using the XM16E1 rifle it is sometimes necessary to hit the "forward assist" with the right hand in order to accomplish full closure of the bolt).

3.2.7 The lanyard is carefully attached to the trigger. If any delay should occur after the cartridge has been chambered, and the duration of the delay is such that the temperature of the cartridge has changed significantly, that cartridge shall be extracted and another inserted in it place.

3.2.8 The technician retires to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer. The fired case is visually examined by the technician for possible case casualties.

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3.2.9 The technician then recovers the fired proof-slug from the recovery container and examines it carefully. If the proof-slug is not damaged in any way, it shall be re-seated on the flash suppressor on the muzzle of the weapon and used again. Should the proof-slug be damaged in any way, it shall be replaced with a new proof-slug.

3.2.10 The procedure prescribed in 3.2.5.1 through 3.2.9 is followed for each cartridge to be fired.

3.2.11 Continuous air cooling shall be used on the barrel. If air cooling is not available, barrel shall be allowed to cool to ambient temperature between each series of tests, or after a maximum number of twenty-five cartridges has been fired.

3.2.12 If any of the personnel conducting the test observes any abnormality that may invalidate the velocity measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, even though the possible cause cannot be immediately assigned, the circumstances should be reported immediately for investigation and decision by proper supervisory authority.

4.0 RECORDING OF DATA

4.1 Individual velocities, average velocity, extreme variation and standard deviation of the individual velocities.

4.2 Number and type of case defects. Casualties shall be reported in accordance with terminology specified in 6.0 and the applicable specification.

4.3 Failures of gun parts shall be shown on the ammunition report and gun card.

4.4 The following test weapon data shall be recorded on the test sheet form:
a. Weapon number and type
b. Total number of cartridges fired in rifle.

5.0 OPERATIONAL NOTES

5.1 In the event any malfunction occurs during firing of the test, a detailed check shall be made to determine whether the ammunition or the equipment is at fault. The check of the weapon shall include measurement of the firing-pin protrusion and firing-pin indent. To assist in determining whether ammunition or equipment is responsible for a malfunction, it is good practice to test the weapon in question using ammunition of known characteristics, and to test the ammunition in question by firing in another weapon of the same type. If it is established that

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some faulty condition of the weapon is responsible for the malfunction, then the test shall be disregarded, the weapon shall be corrected or replaced, the test shall be refired.

5.2 If a misfire is encountered, the weapon is examined carefully to determine if the cause is attributable to the gun. In any ballistic acceptance test where a misfire occurs, a second attempt to fire a primer is not made. It is mandatory that a period of at least five minutes elapse after the misfire occurs before the action of the weapon is opened, where-upon the misfired cartridge is carefully removed in accordance with existing safety regulations, and preserved for further examination. All handling and examination of misfired cartridges shall be conducted with due regard for the hazards involved. The weapon in which a misfire occurs shall be thoroughly checked; it shall be disassembled and all component parts critically scrutinized. Results of such examination shall be included on the test report as a matter of information.

5.2.1 Laboratory examination of the misfired cartridges shall be made to determine the specific cause; the result of the investigation shall be included on the test report.

5.3 Upon completion of firing, all cartridge cases from the test ammunition shall be carefully examined for firing defects. If any defect is found, a detailed check of the equipment shall be made to determine whether the ammunition or the equipment is at fault. If it is established that a faulty weapon is responsible for the firing defect, then the test shall be disregarded, the weapon shall be corrected or replaced, and the tests shall be refired. If it cannot be established that the weapon or other equipment is at fault, then the firing defects shall be charged against the ammunition.

6.0 DEFINITIONS

6.1 Misfire.- Failure of a round of ammunition to fire after initiating action is taken. There are two general categories of misfires:

a. The primer fails to fire when struck by the firing pin.
b. The propellant fails to ignite when the primer fires normally.

6.2 Perforated Primer.- A perforated primer is one in which the indent in the primer cup, made by the firing pin, is entirely perforated. It can be identified by a visible hole through the primer, or, if the perforation be minute, by discoloration of the indent caused by gas burning.

6.3 Primer Leak.- Discoloration caused by gas leakage around the junction between the primer cup and the primer pocket wall.

6.4 Loose Primer.- Looseness, but not so as to permit the fired primer to fall from the primer pocket after the cartridge is fired.

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6.5 Blown Primer, or a primer which falls out of the primer pocket.- A blown primer is a primer which, when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the pocket are enlarged and deformed. A primer which falls out of the pocket is in the same category as a blown primer but the distortion to the primer pocket is less obvious.

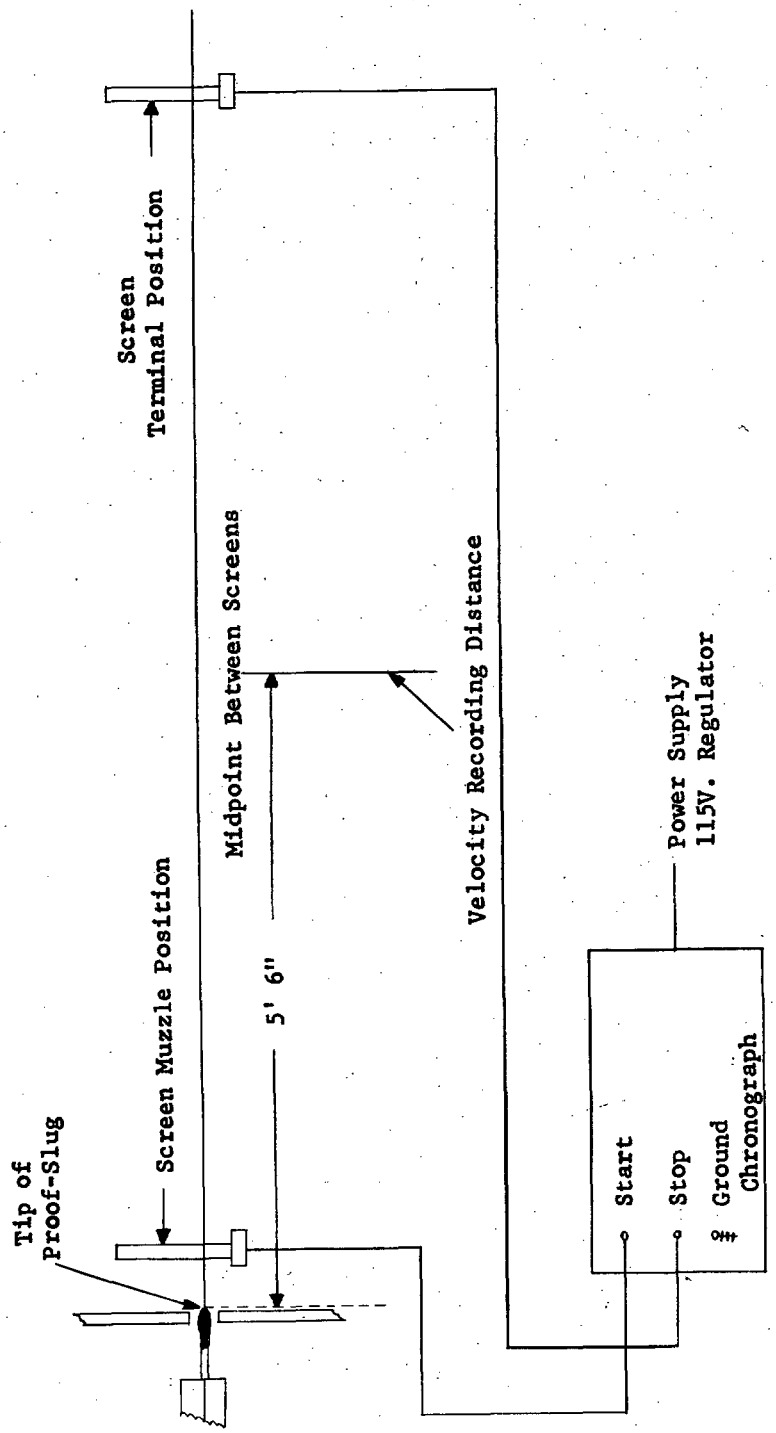
6.6 Ruptured Case.- A circumferential separation of the case wall produced by firing. Ruptures are divided into two categories, partial and complete. A partial rupture is one which extends less than 360° around the case; a complete rupture is one which extends entirely around the case, separating the case into two parts. Ruptures are designated according to position, as indicated on drawing C7643674.

6.7 Split Case.- A longitudinal separation of the metal in the case wall, produced by firing. Splits shall be classified as prescribed by the cartridge specification.

6.8 Failure to Extract.- The fired case is not removed from the weapon chamber by normal gun action.

6.9 Primer Setback.- The backward movement of a primer cup in a cartridge case which occurs when the base of the cup is not properly supported by the bolt face or breech block, and protrudes above the head of the cartridge case.

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RANGE EQUIPMENT, VELOCITY TEST FOR CARTRIDGE GRENADE, RIFLE, 5.56 MM

Chart #1

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Engineering Proof Test Record
Velocity and
Function & Casualty Tests

Propellant:

Type
Army Lot
Charge
Case
Primer Spec/Auth

Ammunition:

Lot
Type
Caliber

Receiver No.	Barrel No.	Times Fired	Firing Pin Protrusion	Firing Pin Indent	Headspace

	Stored at / Fired at		Stored at / Fired at		Stored at / Fired at		Total All Conditions
Shot No.	Velocity	F & C*	Velocity	F & C*	Velocity	F & C*	F & C*
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
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22							
23							
24							
25							
Total							
Mean							
Ex. Var.							
STd. Dev.							

* For legend, see Item Specification, no entry indicates "No Defects".

REMARKS:

Proof Technician(s)
Date Fired

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APPENDIX "B"
Temperature (High & Low) Test Procedure

1.0 PURPOSE

To determine the effect of high or low temperature storage upon the ballistic performance of ammunition or components thereof.

2.0 EQUIPMENT

2.1 Equipment listed in the Temperature test section of the Inspection Equipment List shall be used.

2.2 Controlled-temperature container or containers capable of maintaining temperatures within a tolerance of ± 2°F for temperature of 70°F. (Insulated boxes which have been brought to the required temperature may be used to transfer ammunition from controlled-temperature storage to the test weapons.)

2.3 The controlled-temperature container(s) shall be of sufficient capacity to permit conditioning the ammunition to the desired temperature, and to allow free circulation of air.

3.0 TEST PROCEDURE

3.1 Preparation for test

The same procedure used in preparation for firing Velocity, Function and Casualty (Appendix "A") shall be used for this test with the following exceptions:

3.1.1 Low temperature

The controlled-temperature container shall be cooled to the specified temperature before the test cartridges are placed inside. When the controlled-temperature container indicates the temperature specified, the ammunition shall be placed in the container in such a manner that all cartridges are subjected to a uniform temperature and held thereat for the time specified in the item specification. The weapon in which the ammunition is fired shall be at ambient temperature.

3.1.2 High temperature

The controlled-temperature container shall be heated to the specified temperature before the test cartridges are placed inside. When the controlled-temperature container indicates the temperature specified, the ammunition shall be placed in the container in such a manner that all cartridges are subjected to a uniform temperature and held thereat for the time specified in the item specification. The weapon in which the ammunition is fired shall be at ambient temperature.

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3.1.3 Temperature of test ammunition shall be 60° to 80°F. prior to conditioning at high or low temperature.

3.2 Firing the test

3.2.1 The ammunition under test, stored at the temperature specified in the applicable specification, is removed from the controlled-temperature container and placed in the insulated box (referenced in 2.2), which has also been conditioned to the temperature specified. The insulated box is then placed at a point convenient to the technician.

3.2.2 The procedure used for firing the Velocity, Function and Casualty Test (Appendix "A") shall be used for this test; however, five cartridges are removed from the controlled-temperature container and placed in the insulated box. The cartridges are then removed singly from the insulated box and fired.

4.0 RECORDING OF DATA

4.1 Results shall be recorded directly on the test report form.

4.1.1 Velocity.- Individual velocities and the average velocity, extreme variation, standard deviation of the individual velocities, at each specified temperature.

4.1.2 Function and Casualty.- Number and type of case casualties and weapon malfunction at each specified temperature.

4.1.3 Temperature at which test cartridges were stored and fired.

4.2 The following test weapon data shall be recorded on the test sheet:
a. Weapon number and type.
b. Total number of cartridges fired in weapon prior to test.
c. Headspace measurement.

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